Analytics Vidhya

SageMaker Studio for Training

Instructor

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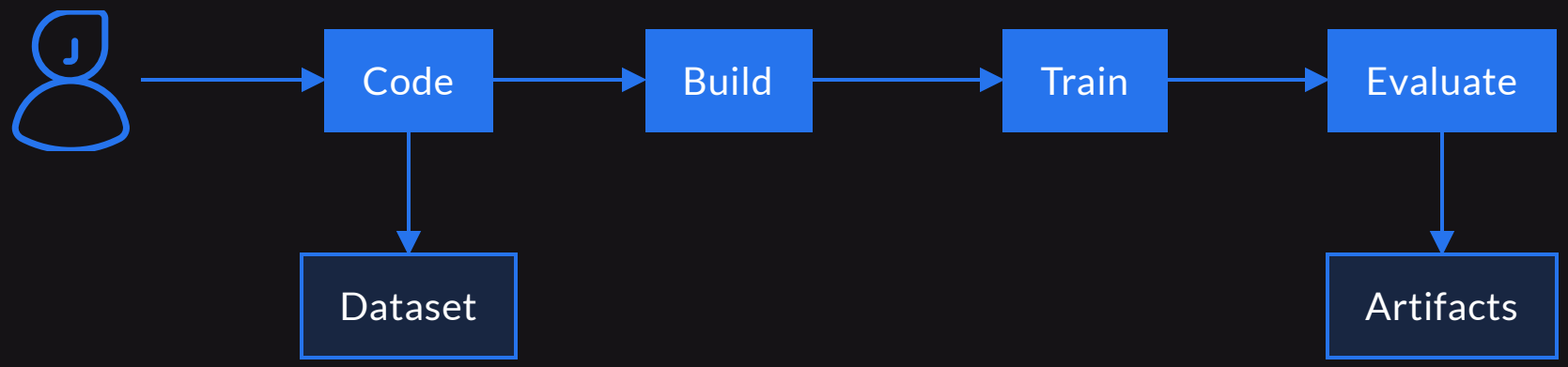
ML Workflow



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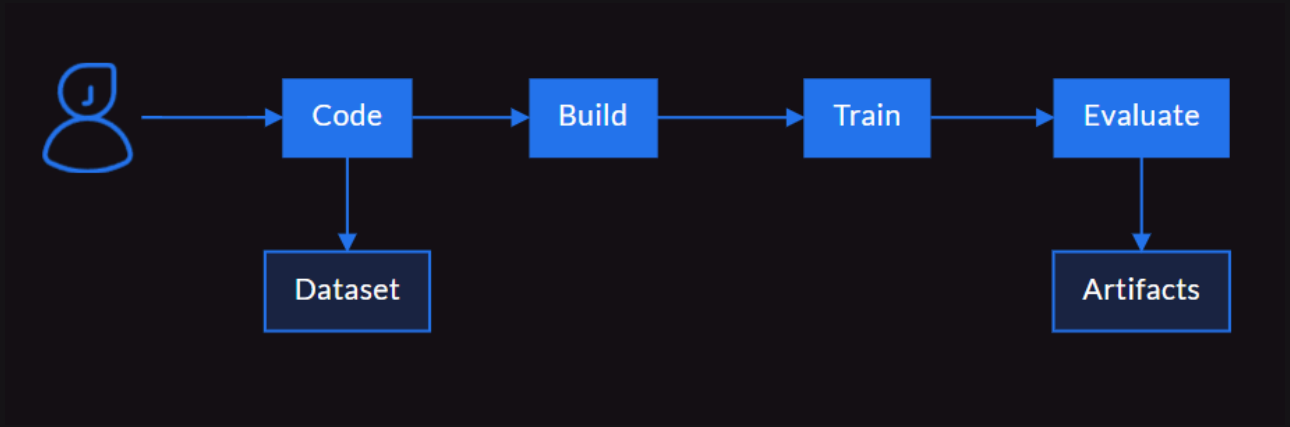
ML Workflow

We need to organize, track, compare and evaluate ML experiments

What needs to be tracked ?

- Dataset
- Data preprocessing code
- Versioned libraries and framework
- Hyperparameters
- Experiment logs
- Training metrics
- Evaluation metrics
- Model weights
- Compare different runs together



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Development Environment

Runtime environments

- Code Editor: VS Code
- JupyterLab
- Canvas
- RStudio
- Studio classic
- MLflow

Integration

- REST APIs
- CLI
- SDK

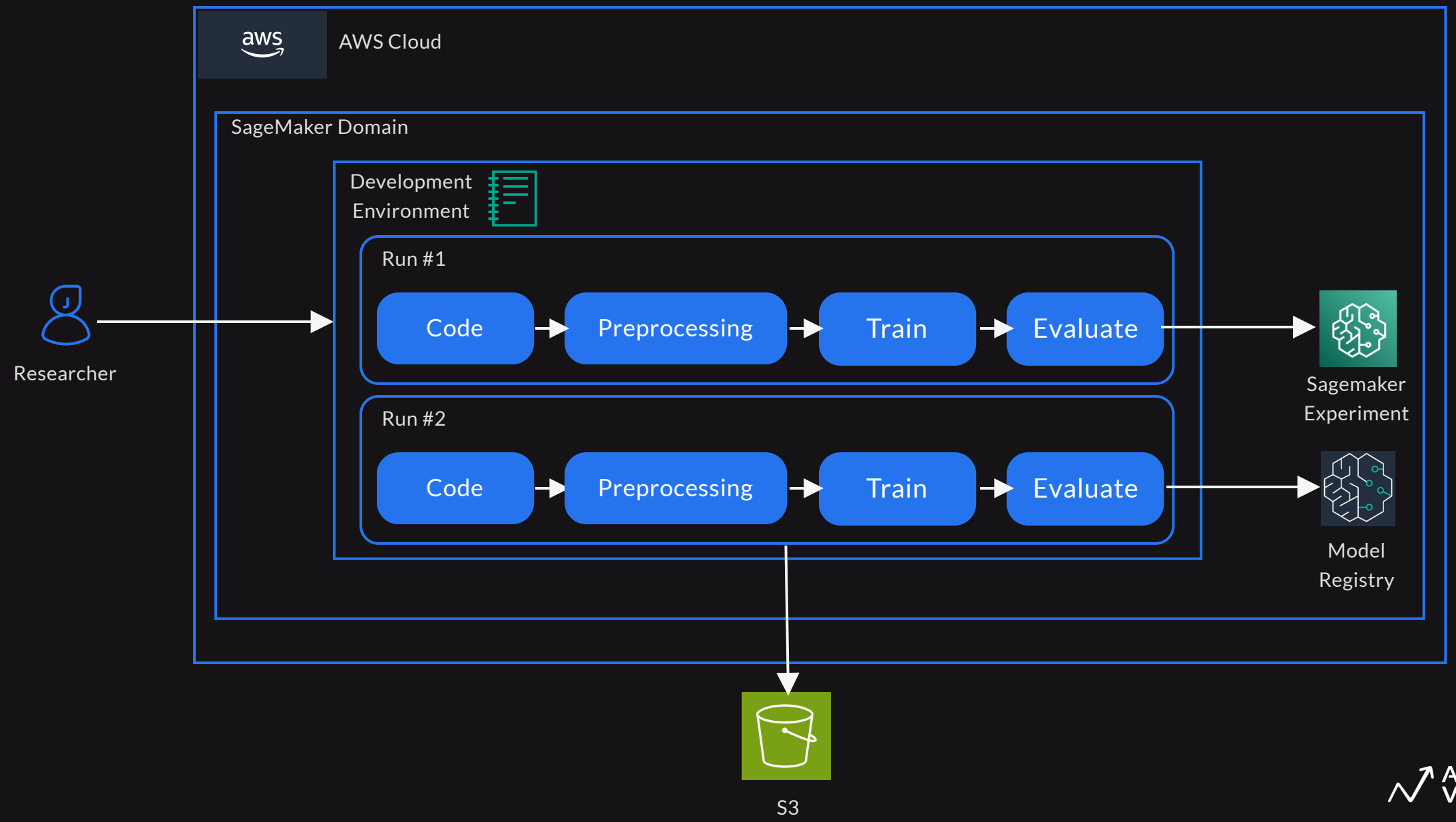
Python SDK (pypi)

- sagemaker
- boto3

Other Services

- SageMaker JumpStart
  - Quickly test Foundation Model
  - Test end-to-end solutions from AWS Marketplace
- Amazon Bedrock
  - Pay as you go FM in AWS regions

SageMaker Experiments



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SageMaker Experiments

- Experiment consists of multiple runs in it
- Run consists of multiple metrics logged and stored in SageMaker experiment
- Deep integration with SDK
- Organized in Studio dashboard UI

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Thank You